Supply Chain and Logistics

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Introduction to Supply Chains

A supply chain is the entire network of people, organisations, resources, activities and information involved in creating a product and delivering it to the final customer. It starts with raw materials and ends with the consumer. For example, the supply chain of a cotton T-shirt begins with farmers growing cotton, then ginners, spinning mills that make yarn, weaving or knitting units that make fabric, dyeing and garment factories, then transporters, warehouses, distributors, retailers or online platforms, and finally the customer who wears it. Tiruppur in Tamil Nadu is a famous example of a garment supply chain in India. Supply chain management (SCM) is the planning, coordination and control of all these activities so that the right product reaches the right place at the right time, at the right cost and quality.

Parts of the Supply Chain

The main stages are sourcing and procurement (buying raw materials from suppliers), manufacturing or production, warehousing and inventory management, transportation and distribution, retail and sales, and returns (reverse logistics). Three flows move through the chain: the physical flow of goods from suppliers to customers, the flow of information such as orders and forecasts, and the flow of money in payments. The players include suppliers, manufacturers, wholesalers, distributors, retailers, third-party logistics providers (3PL), banks and customers.

What Is Logistics?

Logistics is the part of the supply chain that deals with the movement and storage of goods: transport (by road, rail, air, sea and pipeline), warehousing, inventory control, packaging, order processing and delivery. Supply chain management is the broader concept that includes logistics, as well as procurement, production planning, supplier relationships and demand forecasting. Last-mile delivery is the final step to the customer. Cold chain logistics is the temperature-controlled transport of perishable goods such as milk, vegetables, fish, medicines and vaccines. Containers, standardised steel boxes introduced by Malcolm McLean in 1956, made global shipping cheap, and about 90 percent of global trade by volume moves by sea.

What Is Just-in-Time Inventory?

Just-in-Time (JIT) is an inventory and production strategy in which materials and components are ordered and received only when they are needed in the production process, rather than being stored in large quantities in advance. The goal is to reduce the cost of holding inventory, minimise waste, free up cash and storage space, and improve efficiency. JIT was developed by the Japanese company Toyota in the 1950s to 1970s under Taiichi Ohno and Kiichiro Toyoda, as a central part of the Toyota Production System. It uses tools such as kanban, a card-based signalling system, and works best with reliable suppliers, close relationships and accurate demand forecasting. For example, a car factory in Chennai may receive seats, tyres or dashboards from nearby suppliers just a few hours before they are fitted into cars on the assembly line. The advantages are lower costs, less waste, less storage and quicker response to changes. The disadvantage is that JIT leaves no buffer: if a supplier is late or a shipment is disrupted, the whole production line can stop. The opposite approach, holding extra stock for safety, is called just-in-case.

How Did COVID-19 Affect Global Supply Chains?

The COVID-19 pandemic exposed the fragility of global supply chains. In early 2020, lockdowns shut factories in China, the world's manufacturing hub, which disrupted the supply of components and finished goods across the world. As lockdowns spread, workers could not go to factories, ports and warehouses, and migrant workers in India left cities, leading to labour shortages. Demand patterns changed suddenly, with a surge in orders for masks, sanitisers, medical equipment, laptops, home fitness and online shopping, and a drop in demand for travel and other services. Container shipping was thrown into chaos: empty containers were in the wrong places, ports were congested, ships waited for days to unload, and freight rates rose several times, with the cost of shipping a container from China to Europe or the US rising to about ten times its normal level in 2021. A global shortage of semiconductor chips from 2020 to 2023 forced carmakers, including those in Chennai, to cut production. In March 2021, the container ship Ever Given blocked the Suez Canal for six days, holding up trade worth billions of dollars a day. The result was shortages, delays, higher prices and contributions to global inflation. Lessons learned include the need for resilience: businesses began diversifying suppliers, adopting a "China plus one" strategy, shifting production closer to home (nearshoring and reshoring), keeping larger safety stocks, using digital tools for visibility and planning for risks. India promoted self-reliance through the Atmanirbhar Bharat campaign and production-linked incentive schemes.

Logistics in India

India's logistics costs have been estimated at about 13 to 14 percent of GDP, higher than in many developed countries. The government launched the PM Gati Shakti National Master Plan in October 2021 to coordinate infrastructure projects, and the National Logistics Policy in September 2022 aims to reduce costs and improve efficiency. Other initiatives include the Dedicated Freight Corridors for railways, the Sagarmala programme for ports, the Bharatmala programme for highways, FASTag for toll payment, the e-way bill, the Unified Logistics Interface Platform (ULIP) and GST, which removed state check posts. Chennai, with its ports at Chennai and Kamarajar (Ennore) and Kattupalli, a large container terminal at Chennai Port, rail links and the industrial corridor, is an important logistics hub for South India, handling exports of cars, electronics and leather goods. A famous Indian example of an efficient supply chain is the Mumbai dabbawalas, who deliver about 200,000 lunch boxes every day with an extremely low error rate, using a simple colour-coding system. Amul's cooperative milk supply chain, connecting millions of farmers to consumers, is another success story.

Technology in Supply Chains

Technology such as GPS tracking, IoT sensors, barcodes and RFID, warehouse robots, drones, artificial intelligence for demand forecasting, blockchain for traceability and digital twins is making supply chains smarter. Amazon and Flipkart use automated warehouses and algorithms to speed up delivery.

The Bullwhip Effect and Sustainability

The bullwhip effect describes how small changes in customer demand are amplified as they move up the supply chain, causing large swings in orders and inventory. Sharing information reduces this. Sustainable supply chains aim to reduce emissions through electric trucks, rail transport, efficient packing and recycling.

Challenges and Future

Challenges include high costs, infrastructure gaps, geopolitical tension, extreme weather, cyberattacks and labour shortages. The future involves resilient, transparent, digital and green supply chains.

Quick Facts

- A supply chain is the network that moves a product from raw material to the final customer.
- Just-in-Time inventory means receiving materials only when needed to cut inventory costs; Toyota developed it.
- COVID-19 caused factory shutdowns, port congestion, a container shortage and a chip shortage, exposing supply chain fragility.

- PM Gati Shakti was launched in October 2021 and the National Logistics Policy in September 2022.
- India's logistics cost is about 13 to 14 percent of GDP.